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PolyPLASTY PolyPOM Polyoxymethylene (POM)

Categories: [Polymer](#); [Thermoplastic](#); [Acetal \(Polyoxymethylene, POM\)](#); [Acetal Copolymer, Unreinforced](#)

Material Notes: POM has minimal water absorption, excellent dimensional stability and the best properties for chip machining. Thanks to its favorable characteristics, great hardness, rigidity and strength with good toughness, chemical resistance and its shear behavior with abrasion resistance, POM is a successful replacement for metal materials. It is resistant to organic solvents. It also features good resistance to black light UV-radiation.

Information provided by PolyPLASTY.

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

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Physical Properties	Metric	English	Comments
Water Absorption	0.20 %	0.20 %	under regular conditions; ISO 62
Moisture Absorption at Equilibrium	0.80 %	0.80 %	ISO 62
Mechanical Properties	Metric	English	Comments
Hardness, H358/30	145 MPa	21000 psi	ISO 2039-1
Hardness, Shore D	81	81	ISO 868, ISO 2039-2
Elongation at Break	27 %	27 %	ISO 527
Elongation at Yield	9.4 %	9.4 %	ISO 527
Modulus of Elasticity	2.70 GPa	392 ksi	ISO 527
Flexural Strength	97.0 MPa	14100 psi	ISO 178
Fatigue Strength	31.0 MPa	4500 psi	flexure; ASTM D671
Izod Impact, Notched	0.600 J/cm	1.12 ft-lb/in	ISO 180/4A
Izod Impact, Notched (ISO)	5.50 kJ/m ²	2.62 ft-lb/in ²	ISO 180/1A
	@Temperature 23.0 °C	@Temperature 73.4 °F	
	5.50 kJ/m ²	2.62 ft-lb/in ²	ISO 180/1A
	@Temperature -30.0 °C	@Temperature -22.0 °F	
Izod Impact, Unnotched (ISO)	100 kJ/m ²	47.6 ft-lb/in ²	ISO 180/1C
	@Temperature -30.0 °C	@Temperature -22.0 °F	
	930 kJ/m ²	443 ft-lb/in ²	ISO 180/1C
	@Temperature 23.0 °C	@Temperature 73.4 °F	
Charpy Impact Unnotched	19.0 J/cm ²	90.4 ft-lb/in ²	ISO 179/1eU
	@Temperature -30.0 °C	@Temperature -22.0 °F	
	21.0 J/cm ²	99.9 ft-lb/in ²	ISO 179/1eU
	@Temperature 23.0 °C	@Temperature 73.4 °F	
Charpy Impact, Notched	0.550 J/cm ²	2.62 ft-lb/in ²	ISO 179/1eA
	@Temperature -30.0 °C	@Temperature -22.0 °F	
	0.600 J/cm ²	2.86 ft-lb/in ²	ISO 179/1eA
	@Temperature 23.0 °C	@Temperature 73.4 °F	
Tensile Creep Modulus, 1000 hours	1400 MPa	203000 psi	0.5 percent; ISO 899-1

Electrical Properties	Metric	English	Comments
Electrical Resistivity	1.00e+14 ohm-cm	1.00e+14 ohm-cm	IEC 60093
Surface Resistance	1.00e+14 ohm	1.00e+14 ohm	IEC 60093
Dielectric Constant 	3.8	3.8	IEC 60250
	@Frequency 50 Hz	@Frequency 50 Hz	
Dielectric Strength	3.8	3.8	IEC 60250
	@Frequency 1e+6 Hz	@Frequency 1e+6 Hz	
Dissipation Factor 	40.0 kV/mm	1020 kV/in	IEC 60243-1
	@Thickness 1.00 mm	@Thickness 0.0394 in	
Comparative Tracking Index	0.0010	0.0010	IEC 60250
	@Frequency 50 Hz	@Frequency 50 Hz	
	0.0050	0.0050	IEC 60250
	@Frequency 1e+6 Hz	@Frequency 1e+6 Hz	
Thermal Properties	Metric	English	Comments
CTE, linear, Transverse to Flow	110 µm/m-°C	61.1 µin/in-°F	ISO 11359
Specific Heat Capacity	1.47 J/g-°C	0.351 BTU/lb-°F	IEC 1006
Thermal Conductivity	0.310 W/m-K	2.15 BTU-in/hr-ft²-°F	DIN 52 612
Maximum Service Temperature, Air	90.5 °C	195 °F	long term
	140 °C	284 °F	short term
Vicat Softening Point	150 °C	302 °F	point B; ISO 306 VST/B/50
	@Load 5.10 kg	@Load 11.2 lb	
Minimum Service Temperature, Air	-50.0 °C	-58.0 °F	
Flammability, UL94	HB	HB	IEC 60695-11-10
	@Thickness 0.800 mm	@Thickness 0.0315 in	

Descriptive Properties

LOI Acid Index (%)	15	ISO 4589
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